
What Problem Does This Paper Address?

The paper discusses the major causes of motorcycle accidents and how smart helmet technology can help solve them.

Problems Identified

1. Riders do not wear helmets.

Many riders avoid wearing helmets, leading to severe head injuries during accidents.

2. Drunk Driving.

Riding after consuming alcohol causes a large number of road accidents.

3. Over-Speeding.

Many riders exceed speed limits, increasing accident risk.

4. Drowsy Riding.

Sleepy or tired riders may lose concentration and cause accidents.

5. Delayed Emergency Response.

Victims often do not receive medical assistance quickly because nobody knows their location.

6. Poor Communication.

Riders cannot easily communicate with nearby riders or emergency contacts during emergencies.

These issues are introduced as the motivation for smart helmet research.

Main Objective of Smart Ride Helmet

According to the paper, the ideal Smart Ride Helmet should:

- Make helmet wearing compulsory.
- Prevent drunk driving.
- Monitor vehicle speed.
- Detect if the rider becomes sleepy.
- Alert guardians during over-speeding.
- Detect accidents automatically.
- Share GPS location after accidents.
- Allow communication between helmets.
- Reduce road accidents.

These objectives are explicitly stated in the paper.

Features Discussed in the Literature Review

The paper reviews many research projects. Combining their ideas, it identifies the following major smart helmet features.

1. Helmet Detection

Problem

Many riders ignore helmets.

Solution

Sensors such as:

- IR Sensor
- Force Sensor
- Pressure Sensor

detect whether the helmet is being worn.

If helmet is not worn:

- Bike ignition remains OFF.

Benefit

- Makes helmet usage compulsory.
 - Reduces fatal head injuries.
-

2. Alcohol Detection

Problem

Drunk driving causes serious accidents.

Solution

MQ3 Alcohol Sensor checks the rider's breath.

If alcohol is detected:

- Bike ignition is disabled.
- Alert is generated.

Benefit

- Prevents drunk driving.
-

3. Accident Detection

Smart helmets use:

- Accelerometer
- Gyroscope
- Vibration Sensor

to detect accidents.

If an accident occurs:

- GPS gets the location.
- GSM sends an emergency message.

Benefit

- Faster emergency response.
 - Saves lives.
-

4. GPS Location Tracking

GPS continuously tracks the rider.

If an accident occurs:

- Exact location is sent to family members, hospitals, or emergency services.

Benefit

- Faster rescue.
-

5. GSM Emergency Alert

GSM module automatically sends:

- SMS
- Rider location
- Emergency alert

to registered contacts.

Benefit

- Immediate medical assistance.
-

6. Speed Monitoring

Some reviewed helmets monitor vehicle speed.

If speed exceeds the limit:

- Rider receives an alert.
- Guardian can also be notified.

Benefit

- Prevents over-speeding accidents.
-

7. Sleep Detection

The paper mentions detecting whether the rider is sleepy.

If drowsiness is detected:

- Helmet warns the rider.

Benefit

- Prevents accidents caused by fatigue.
-

8. Bluetooth Communication

Some smart helmets include Bluetooth.

This allows:

- Hands-free calling.
- Rider communication.

Benefit

- Safer communication while riding.
-

9. IoT Monitoring

Several reviewed projects use IoT.

IoT allows:

- Live monitoring.
- Remote tracking.
- Cloud storage.
- Real-time alerts.

Benefit

- Better rider safety.
-

10. Theft Protection

Some smart helmets integrate security features.

If:

- Helmet moves far from the bike,

then:

- Engine turns OFF.
- SMS is sent to the owner.

Benefit

- Prevents bike theft.
-

11. Mining Safety

The paper also reviews smart helmets for miners.

These helmets monitor:

- Methane gas
- Carbon monoxide
- Temperature
- Humidity
- Worker health

using IoT and LoRaWAN.

Benefit

- Improves underground worker safety.
-

Technologies Mentioned

The paper summarizes the technologies used across different smart helmet projects.

Technology	Purpose
Arduino / ESP32 / Raspberry Pi	Main controller
IoT	Real-time monitoring
GPS	Rider location
GSM	SMS alerts
MQ3 Sensor	Alcohol detection
IR Sensor	Helmet detection
Accelerometer	Accident detection
Gyroscope	Motion detection
RF Module	Helmet-bike communication
Bluetooth	Hands-free calling

LoRaWAN

Long-range communication

Cloud Computing

Remote monitoring

The comparison table on page 5 summarizes these technologies and their uses.

Advantages

Based on the literature review, smart helmets can:

- Make helmet usage mandatory.
- Prevent drunk driving.
- Reduce over-speeding.
- Detect accidents automatically.
- Send emergency alerts.
- Share GPS location.
- Improve rider safety.
- Reduce accident fatalities.
- Enable IoT-based real-time monitoring.
- Support mining and industrial safety.

These are the common benefits identified across the reviewed studies.

Future Scope

The paper suggests future smart helmets should include:

- AI-based accident prediction.
- Better IoT connectivity.
- Cloud-based monitoring.
- Advanced rider health monitoring.
- Smart communication between helmets.
- Improved theft protection.
- Multi-domain applications (motorcycles, mining, firefighting, industrial safety).

- More intelligent sensors and automation.

These directions are based on the trends observed across the reviewed research papers.

Final Summary

Unlike the previous papers, this document **does not introduce a new smart helmet design**. Instead, it reviews and compares multiple smart helmet systems developed by other researchers. The review concludes that integrating **IoT, GPS, GSM, sensors, microcontrollers, cloud computing, and wireless communication** can significantly improve motorcycle safety by enforcing helmet usage, preventing drunk driving, monitoring speed, detecting accidents, sending emergency alerts, and enabling real-time monitoring. It also highlights future opportunities such as AI, cloud-based analytics, and smart communication to make helmets even more intelligent and effective.